

Session 16 (AIML) -Assignment Question

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- **College Name**: Zeal Polytechnic , Narhe
- **Branch Name**: AIML
- **Github url**: https://github.com/shrutidarwatkar-sketch/Task_ITR.git

Q1. Load the insurance dataset using pandas and display the first 10 rows of the data.

Program screenshot:

```
from google.colab import files
uploaded = files.upload()
```

... Choose files insurance.csv
insurance.csv(text/csv) - 50264 bytes, last modified: 09/07/2026 - 100% done
Saving insurance.csv to insurance (1).csv

```
import pandas as pd
insurance = pd.read_csv("insurance.csv")
insurance.head(10)
```

...

	age	sex	bmi	children	smoker	region	expenses
0	19	female	27.9	0	yes	southwest	16884.92
1	18	male	33.8	1	no	southeast	1725.55
2	28	male	33.0	3	no	southeast	4449.46
3	33	male	22.7	0	no	northwest	21984.47
4	32	male	28.9	0	no	northwest	3866.86
5	31	female	25.7	0	no	southeast	3756.62
6	46	female	33.4	1	no	southeast	8240.59
7	37	female	27.7	3	no	northwest	7281.51
8	37	male	29.8	2	no	northeast	6406.41
9	60	female	25.8	0	no	northwest	28923.14

Q2. Use pandas functions to get basic information about the dataset:

- Shape of the data (shape)
- Column names and data types (info())
- Statistical summary of numerical columns (describe())

Program screenshot:

```
import pandas as pd
#load the dataset
insurance = pd.read_csv("insurance.csv")

#1.shape of the dataset
print("shape of the dataset")
print(insurance.shape)

#2. column names and data types
print("\n dataset information")
insurance.info()

#3. Statistical summary of numerical columns
print("\n statistical summary")
print(insurance.describe())
```

```
... shape of the dataset
(1338, 7)
```

```
dataset information
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   age         1338 non-null   int64
 1   sex         1338 non-null   object
 2   bmi         1338 non-null   float64
 3   children    1338 non-null   int64
 4   smoker      1338 non-null   object
 5   region      1338 non-null   object
 6   expenses    1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

```
statistical summary
      age      bmi  children  expenses
count 1338.000000 1338.000000 1338.000000 1338.000000
mean   39.207025   30.665471   1.094918 13270.422414
std    14.049960    6.098382   1.205493 12110.011240
min    18.000000   16.000000   0.000000 1121.870000
25%    27.000000   26.300000   0.000000 4740.287500
50%    39.000000   30.400000   1.000000 9382.030000
75%    51.000000   34.700000   2.000000 16639.915000
max    64.000000   53.100000   5.000000 63770.430000
```

Q3. Check if the dataset contains any null/missing values. Show the count of null values for each column.

Program screenshot:

```
#check missing values
print("missing values in each column:")
print(insurance.isnull().sum())

... missing values in each column:
age      0
sex      0
bmi      0
children 0
smoker   0
region   0
expenses 0
dtype: int64
```

Q4. Identify numerical and categorical columns in the dataset. Print them separately.

Program screenshot:

```
# Numerical cols
numerical_cols = insurance.select_dtypes(include=['int64', 'float64']).columns
print("Numerical columns:")
print(numerical_cols)

# Categorical columns
categorical_cols = insurance.select_dtypes(include=['object']).columns
print("\n Categorical columns:")
print(categorical_cols)

... Numerical columns:
Index(['age', 'bmi', 'children', 'expenses'], dtype='object')

Categorical columns:
Index(['sex', 'smoker', 'region'], dtype='object')
```

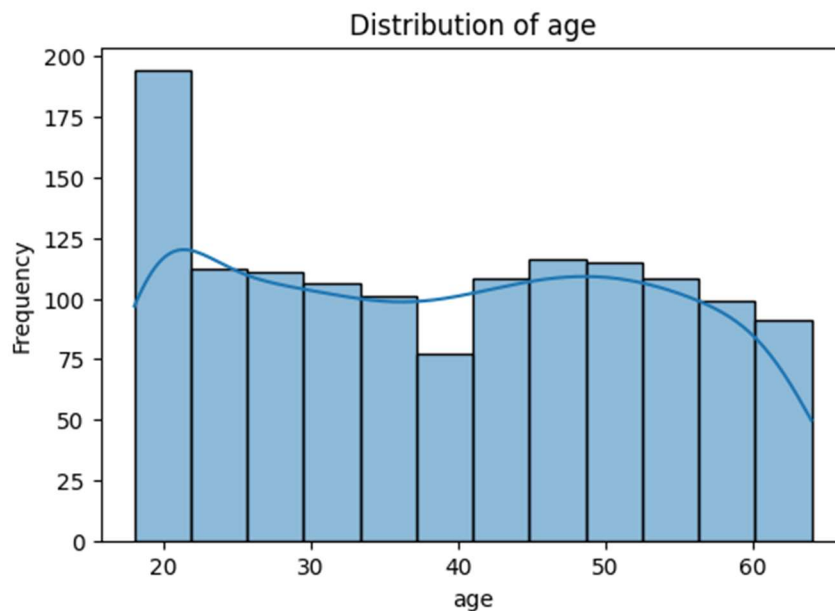
Q5. . Create distribution plots for all numerical columns (age, bmi, children, charges):

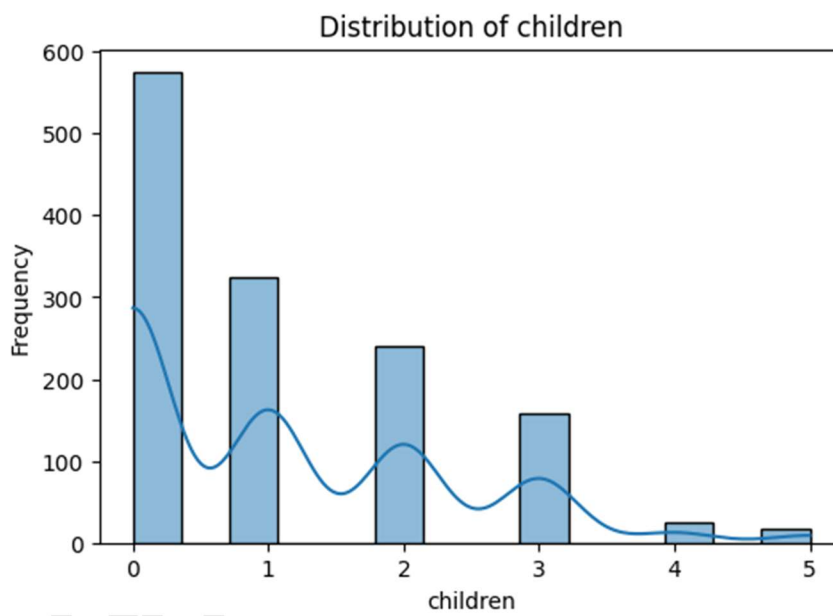
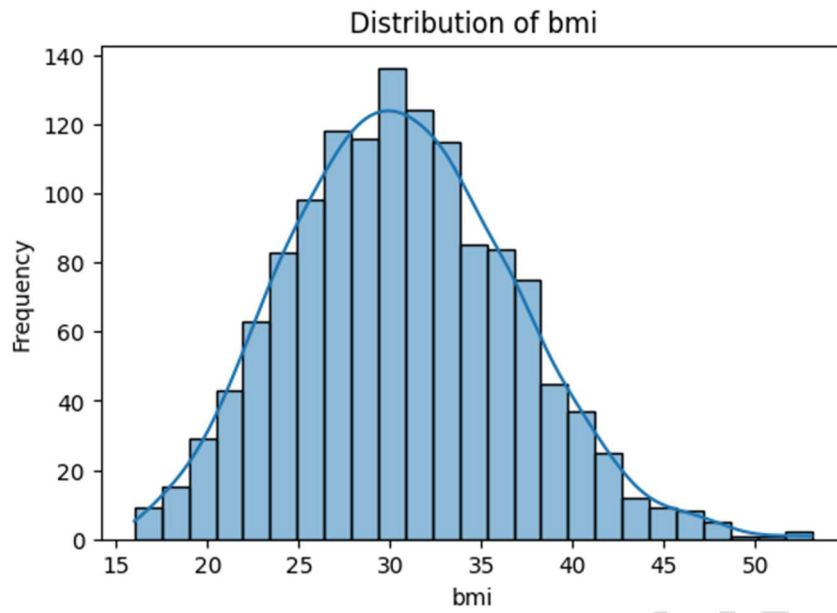
Program screenshot:

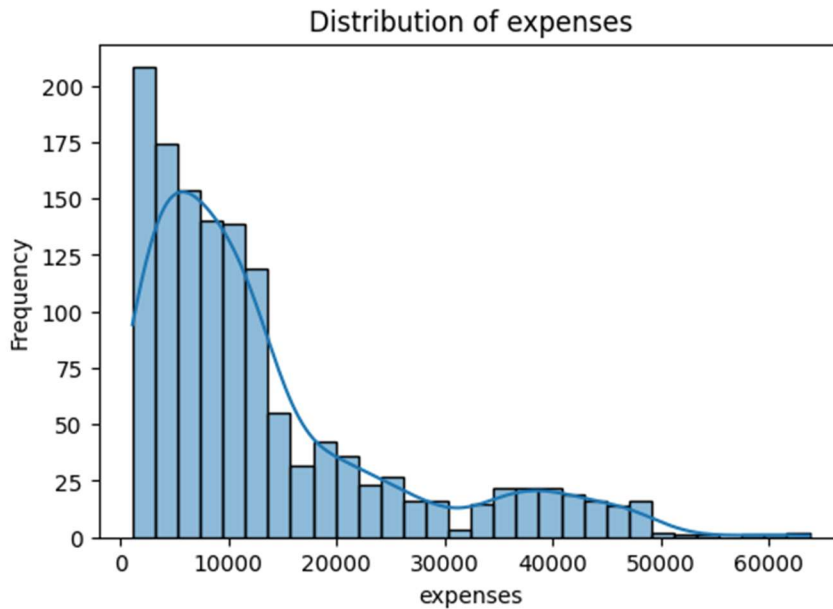
```
import matplotlib.pyplot as plt
import seaborn as sns

# List of numerical columns
num_cols = ['age', 'bmi', 'children', 'charges']
for col in num_cols:
    plt.figure(figsize=(6,4))
    sns.histplot(insurance[col], kde=True)
    plt.title(f"Distribution of {col}")
    plt.xlabel(col)
    plt.ylabel("Frequency")
    plt.show()
```

...







Q6. Create count plots for all categorical columns (sex, smoker, region):

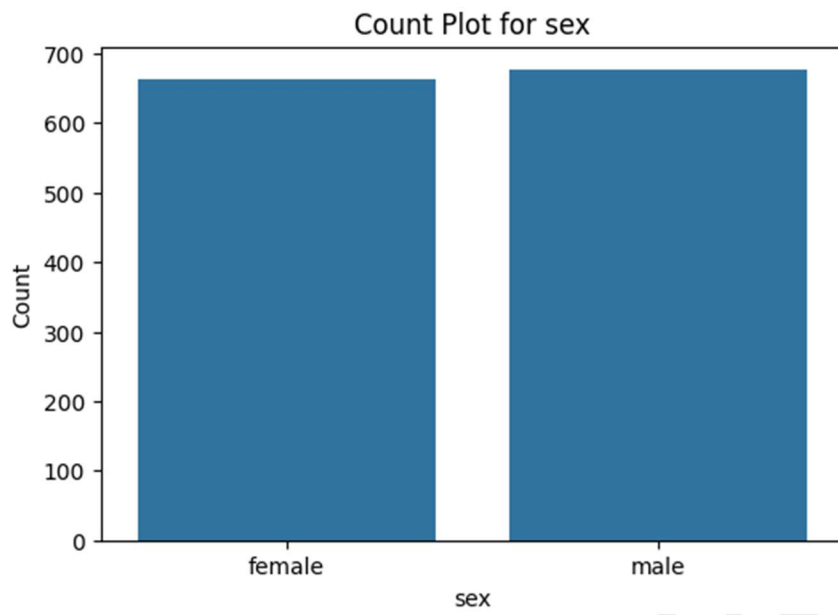
- Use `sns.countplot()` for each column
- Add titles and labels for clarity

Program screenshot:

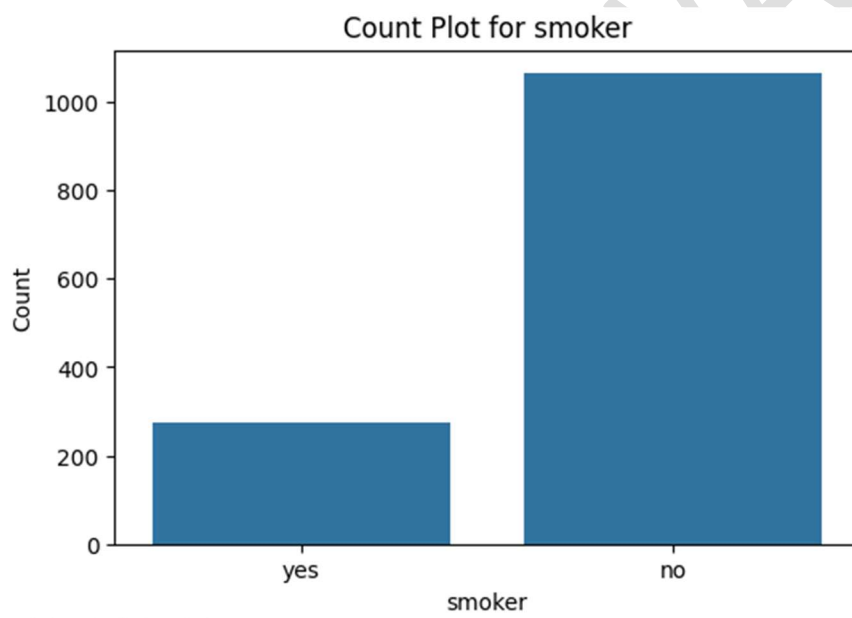
```
#List of categorical columns
cat_cols = ['sex', 'smoker', 'region']

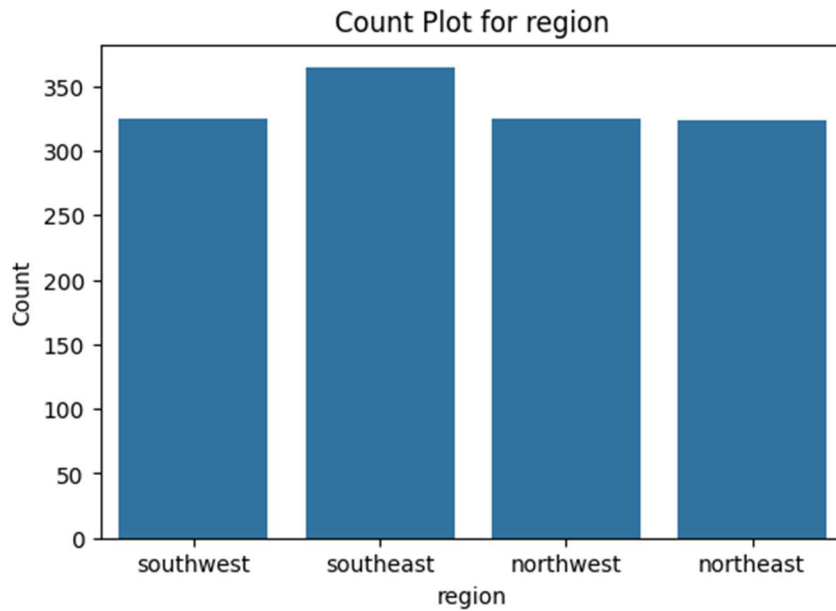
for col in cat_cols:
    plt.figure(figsize=(6,4))
    sns.countplot(x=insurance[col])
    plt.title(f"Count Plot for {col}")
    plt.xlabel(col)
    plt.ylabel("Count")
    plt.show()
```

...



...





Q7. Create a heatmap to show the correlation between numerical variables.

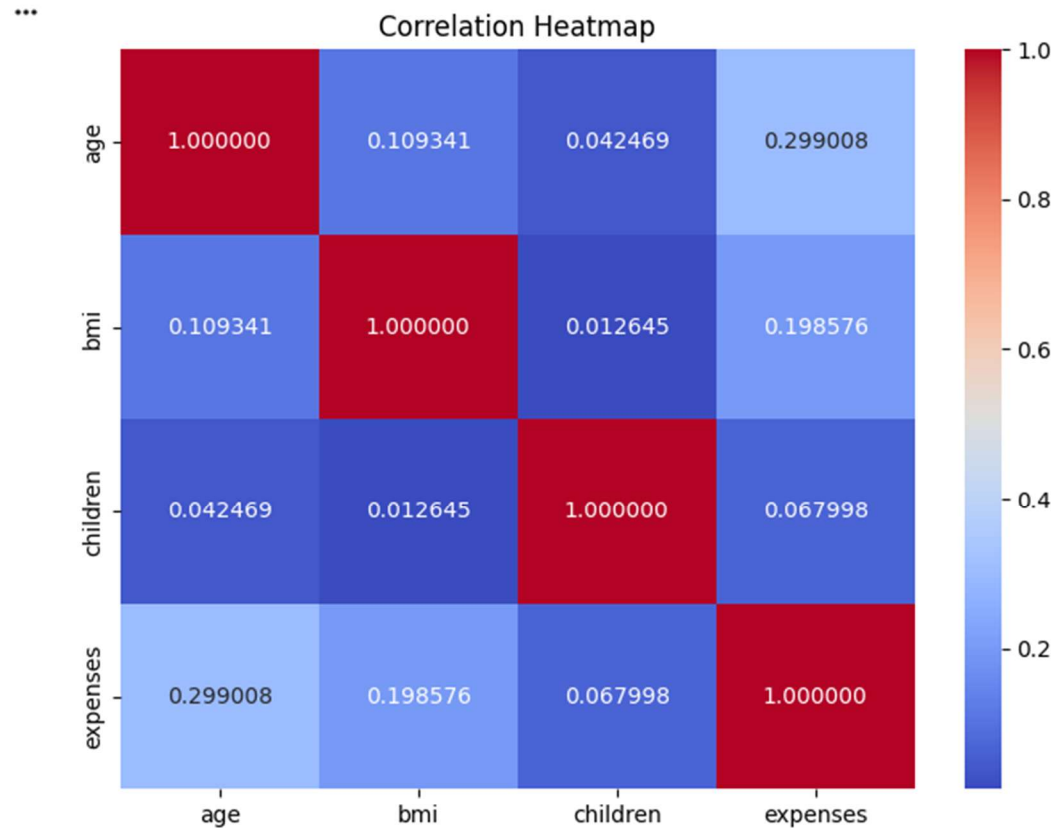
- Use `sns.heatmap()`
- Annotate the values and use a suitable color palette

Program screenshot:

```

plt.figure(figsize=(8,6))
corr = insurance.select_dtypes(include=['int64' , 'float64']).corr()
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt='2f')
plt.title("Correlation Heatmap")
plt.show()

```



Q8. Perform the following analysis on the charges column:

- Find the average insurance charges
- Find the maximum and minimum charges
- Group by smoker and compare the average charges for smokers vs nonsmokers

Program screenshot:

```

# Find the target column (charges/expenses/etc.)
target = None

if 'charges' in insurance.columns:
    target = 'charges'
elif 'expenses' in insurance.columns:
    target = 'expenses'
else:
    # Last numerical column
    target = numerical_cols[-1]

print("\nTarget Column:", target)

print("\nAverage:")
print(insurance[target].mean())

print("\nMaximum:")
print(insurance[target].max())

print("\nMinimum:")
print(insurance[target].min())

if 'smoker' in insurance.columns:
    print("\nAverage by Smoker:")
    print(insurance.groupby('smoker')[target].mean())

```

```

...
Target Column: expenses

Average:
13270.422414050823

Maximum:
63770.43

Minimum:
1121.87

Average by Smoker:
smoker
no      8434.268449
yes     32050.231971
Name: expenses, dtype: float64

```

Q9. Create a boxplot to show the distribution of charges with respect to smoker and sex.

(You can create two separate boxplots or use hue parameter)

Program screenshot:

```

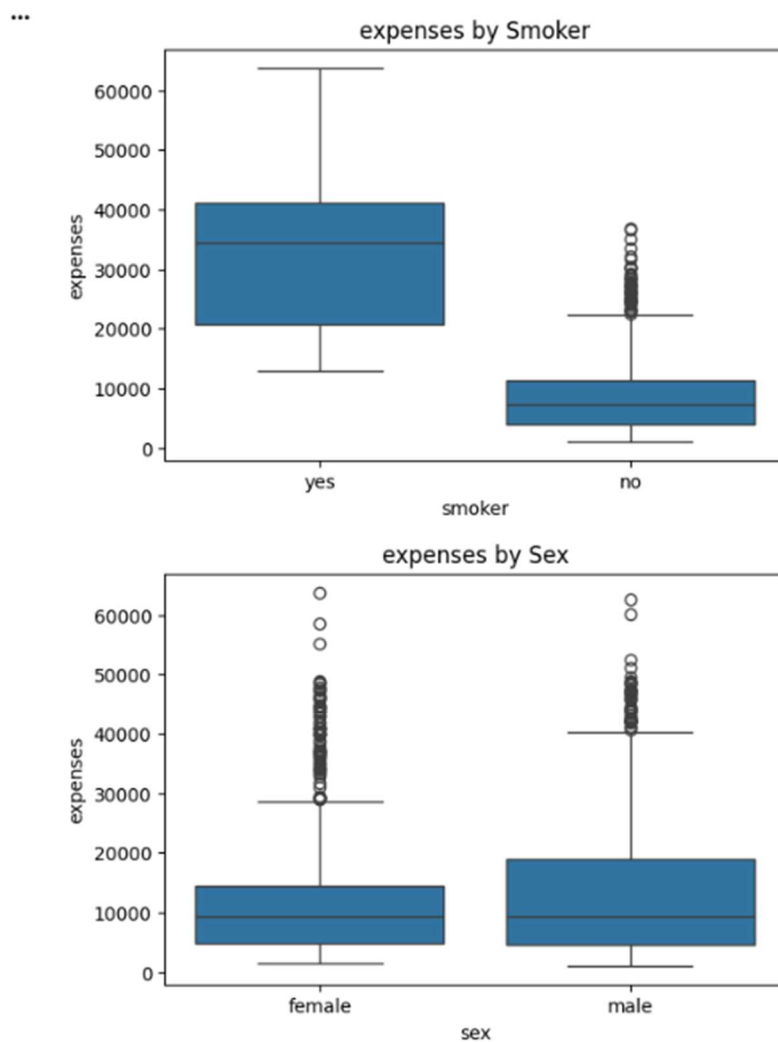
import matplotlib.pyplot as plt
import seaborn as sns

target = insurance.select_dtypes(include=['int64', 'float64']).columns[-1]

plt.figure(figsize=(6,4))
sns.boxplot(x='smoker', y=target, data=insurance)
plt.title(f"{target} by Smoker")
plt.show()

plt.figure(figsize=(6,4))
sns.boxplot(x='sex', y=target, data=insurance)
plt.title(f"{target} by Sex")
plt.show()

```



Q10. Mini Project / Analysis Summary)

1. What is the average age and BMI of the customers?
2. How does smoking affect insurance charges?
3. Which region has the highest number of customers?
4. Any interesting observations from the visualizations?

```
# Mini Project - Analysis Summary

# 1. Average age and BMI are calculated using:
#   insurance['age'].mean()
#   insurance['bmi'].mean()

# 2. Smokers have significantly higher insurance charges
#   compared to non-smokers.

# 3. The region with the highest number of customers is obtained using:
#   insurance['region'].value_counts().idxmax()

# 4. Observations:
#   - Charges are right-skewed.
#   - Most BMI values are between 25 and 35.
#   - Male and female customers are nearly equal in number.
#   - Smokers pay much higher insurance charges.
#   - Customer distribution across regions is fairly balanced.
```